Tertiary Infotech Academy Pte Ltd

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LEARNER GUIDE

For

ISO 14064 Internal Auditor Training

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1.0	Legacy release	Original trainer slide deck.	Tertiary Infotech Academy
2.0	19 March 2026	Prior Learner Guide and Lesson Plan release on the course Drive.	Tertiary Infotech Academy
3.0	20 August 2026	Complete source-aligned visual rebuild; expanded ISO 14064 audit, lifecycle, quantification, uncertainty and benchmarking coverage; ten connected activities and aligned WA/CS assessments.	Dr Alfred Ang

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TABLE OF CONTENTS

How to Use This Guide.....4
Course Learning Outcomes.....4
Competency Alignment.....4
Current Standards Baseline.....5
Topic 01 — Introduction to ISO 14064 and Verification Principles.....5
Topic 02 — GHG Sources, Boundaries and Mitigation Activities.....13
Topic 03 — Lifecycle Assessment and Carbon Footprint Quantification.....21
Topic 04 — Emission Calculations, Uncertainty and Performance Evaluation.....28
Assessment Preparation.....35
Sources and Further Reading.....35

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How to Use This Guide

The slide deck is concept-led. This guide carries the detailed procedures, scenarios, questions, evidence requirements and acceptance checks for all ten activities. Each activity also has a self-contained folder under activities/ with a working Excel workbook and printable PDF/checklist.

Use only fictional, anonymised or authorised workplace information. Apply confidentiality, due professional care, transparent calculation and evidence-based judgement throughout.

Course Learning Outcomes

- LO1: Develop process mappings and document GHG accounting reports aligned to ISO 14064 and international verification guidelines.
- LO2: Define lifecycle boundaries and determine emission sources to identify carbon reduction opportunities in line with assessment principles.
- LO3: Construct lifecycle inventory data for carbon footprint assessments using product configurations and scoped emission categories.
- LO4: Determine uncertainty factors and evaluate product carbon performance against industry benchmarks.

Competency Alignment

Code	Competency requirement
K1	ISO 14001, ISO 14064 and other relevant standards and guidelines
K2	Concepts and principles of carbon footprint assessments
K3	Product lifecycles and product configurations
K4	Types of emissions and corresponding sources
K5	Scopes of carbon footprints
K6	Types of uncertainty factors
K7	Industry benchmarks for specific products' carbon footprints
K8	International guidelines and principles for carbon footprint assessments and documentation
A1	Develop process mappings in accordance with product lifecycle requirements
A2	Define boundary conditions, assumptions and prioritisation for product lifecycles according to international guidelines and principles

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Code	Competency requirement
A3	Construct lifecycle inventory data based on developed product lifecycle boundaries
A4	Assess carbon footprints of product lifecycles using carbon footprint assessment software tools
A5	Determine uncertainty factors based on carbon footprint assessments
A6	Evaluate carbon footprint performance of selected products against industry benchmarks
A7	Determine emissions hot spots and propose potential carbon footprint reduction approaches
A8	Document results and prepare reports according to international guidelines and principles

Current Standards Baseline

- ISO 14064-1:2018 specifies organisation-level principles and requirements for designing, developing, managing and reporting GHG inventories.
- ISO 14064-2 addresses project-level emission reductions and removal enhancements; ISO 14064-3 addresses validation and verification of GHG statements.
- The 2018 revision uses a significance-based approach to indirect emissions and organises reporting into direct emissions plus five indirect-emission categories.
- Confirm the authorised standard text, programme rules, legal obligations, factor/GWP sources and reporting-period requirements before a formal conformance or public claim.

Topic 01 — Introduction to ISO 14064 and Verification Principles

Alignment: K1 · K2 · K8 · A1 · A8 · LO1

ISO 14064 family · ISO 14001 integration · accounting principles · audit and verification practice · reporting evidence

ISO 14064 family and assurance architecture

- **Part 1:** Organisation-level principles and requirements for inventory design, quantification and reporting.
- **Part 2:** Project-level quantification, monitoring and reporting of emission reductions or removal enhancements.
- **Part 3:** Requirements and guidance for validating and verifying GHG statements.

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- **Internal audit:** An organisation-led evaluation of readiness, controls and evidence; it does not itself create third-party assurance.

Auditor application

- Name the applicable part and edition
- Separate organisation, project and product assertions
- Define intended users and assurance purpose
- Use authorised standards text for conformance

Judgement cautions

- ISO 14064 is not one undifferentiated certificate
- A project credit is not an organisational removal
- Internal audit does not replace independent verification
- Do not claim conformance from a checklist alone

Purpose and architecture of ISO 14064-1:2018

- **Inventory design:** Set purpose, organisational boundary, reporting boundary, base year and methods.
- **Development:** Identify sources and sinks, select data and factors, quantify and consolidate.
- **Management:** Assign roles, maintain information systems, assess uncertainty and review changes.
- **Reporting:** Disclose boundaries, categories, methods, exclusions, uncertainty and performance transparently.

Auditor application

- Trace each report claim to a controlled process
- Test the significance approach for indirect emissions
- Check base-year and recalculation rules
- Confirm management responsibility for the statement

Judgement cautions

- The 2018 edition is not a simple Scope 1-2-3 checklist
- Completeness still requires justified exclusions
- Verification readiness is not verified status
- A polished report can hide weak source controls

ISO 14001 integration and the management-system link

- **Context:** Climate, stakeholder and compliance issues influence EMS and GHG priorities.
- **Planning:** Environmental aspects, obligations, risks and objectives can incorporate GHG sources and targets.

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- **Support and operation:** Competence, communication, document control and operational controls support inventory reliability.
- **Evaluation and improvement:** Monitoring, internal audit, management review and corrective action create a repeatable control cycle.

Auditor application

- Reuse governance without merging criteria
- Map GHG data owners into EMS roles
- Connect findings to corrective action
- Escalate material climate and claims risks

Judgement cautions

- ISO 14001 certification does not prove inventory accuracy
- GHG boundaries can differ from EMS boundaries
- One audit plan may not cover both standards fully
- Avoid duplicate uncontrolled records

Related standards and frameworks

- **ISO 14065 and 14066:** Requirements for validation/verification bodies and competence of GHG teams.
- **ISO 14067:** Product carbon-footprint principles and requirements within lifecycle assessment.
- **ISO 19011:** Guidance for audit programmes, risk-based planning, evidence and auditor competence.
- **GHG Protocol:** Widely used corporate, Scope 2 and value-chain accounting guidance that can complement declared ISO criteria.

Auditor application

- Create a criteria crosswalk
- Record definitions that differ
- Use programme rules when contractually applicable
- Keep the controlling criteria explicit

Judgement cautions

- Related does not mean interchangeable
- Scope labels and ISO categories are not identical
- Product and organisation footprints answer different questions
- A crosswalk needs expert judgement

GHG accounting principles as audit tests

- **Relevance:** Boundaries and information serve intended users and decisions.

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- **Completeness:** All significant sources and sinks are included or exclusions are justified.
- **Consistency:** Methods support comparison over time and changes are disclosed.
- **Accuracy and transparency:** Bias and uncertainty are reduced and assumptions, evidence and limitations remain visible.

Auditor application

- Turn every principle into an audit question
- Use principles when prescriptive rules are silent
- Seek evidence of trade-off decisions
- Apply professional scepticism to favourable claims

Judgement cautions

- Principles interact and can conflict
- Completeness does not mean perfect data
- Consistency does not freeze weak methods
- Transparency does not cure avoidable error

Verification planning: objectives, scope, criteria and assurance

- **Objectives:** State what the engagement must conclude and for whom.
- **Scope:** Define statement, entity, locations, period, categories and activities covered.
- **Criteria:** Name the standard, programme rules, calculation policies and materiality basis.
- **Assurance:** Limited and reasonable assurance require different evidence depth and conclusion forms.

Auditor application

- Resolve ambiguous criteria before testing
- Link risk assessment to evidence effort
- Document changes to scope
- Align competence and resources to complexity

Judgement cautions

- More samples do not fix unclear criteria
- Materiality is not only a percentage
- Limited assurance is not low quality
- Internal audit should not mimic an external opinion

Risk-based internal audit process

- **Programme:** Prioritise audits by significance, change, prior findings and stakeholder risk.

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- **Plan:** Set objectives, scope, criteria, team, independence, timetable and sampling strategy.
- **Conduct:** Open, interview, observe, inspect, recalculate, corroborate and communicate emerging issues.
- **Close and follow up:** Report findings, agree responsibilities, verify correction and test effectiveness.

Auditor application

- Plan backward from required conclusions
- Maintain independence from inventory preparation
- Use daily team calibration
- Track findings to verified closure

Judgement cautions

- An audit is not an inspection tour
- Interview statements need corroboration
- Prescribing solutions can compromise independence
- Closing a task is not proving effectiveness

Objective evidence, sampling and reporting

- **Evidence types:** Records, observations, interviews, confirmations, analytics and reperformance provide different strengths.
- **Sampling:** Select high-value, high-risk, unusual and representative items using a documented rationale.
- **Triangulation:** Corroborate important conclusions across independent evidence sources.
- **Report trail:** Link criterion, evidence, evaluation, finding, owner, correction and follow-up status.

Auditor application

- Trace top-down and bottom-up
- Record population and sample logic
- Retest corrected calculations
- Distinguish fact, judgement and recommendation

Judgement cautions

- A large sample can still be biased
- One document may be internally consistent but false
- Absence of evidence is not always evidence of absence
- Avoid vague findings such as documentation weak

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Activity 1 — Map the GHG System and Audit Criteria

Alignment: K1 · K2 · K8 · A1 · A8 · LO1 · Duration: 45 minutes

Objective: Translate ISO 14064 and related criteria into a process map and auditable requirements register.

Scenario

HarbourLight Electronics Pte Ltd manufactures high-efficiency LED panels in Tuas. Its 2025 GHG report was assembled by Sustainability from utility bills, procurement spreadsheets and supplier estimates. The company has ISO 14001 certification, but GHG responsibilities, data hand-offs and review approvals are not mapped. A major customer now requests a verification-ready organisational inventory and product-footprint evidence for the Lumina 400 LED panel.

Materials

- HarbourLight organisation and process facts
- ISO 14064 family and related-framework summaries
- Draft GHG report contents
- Process-map and criteria workbook

Activity Questions

- Which ISO 14064 parts apply to each assertion?
- How does ISO 14001 support but not replace GHG criteria?
- Where does evidence enter, change and receive approval?
- Which report sections need a traceable owner and criterion?

Detailed Procedure

1. Read the fact pattern and separate organisational-inventory, product-footprint and verification assertions.
2. List intended users, decisions, reporting period and desired assurance readiness for the audit.
3. Map the end-to-end GHG process from source activity through factor selection, calculation, review, approval and disclosure.
4. Identify roles and evidence hand-offs among Facilities, Finance, Procurement, Logistics, Product Engineering and Sustainability.
5. Build a criteria crosswalk for ISO 14064-1, relevant ISO 14001 controls, ISO 14064-3 concepts and declared GHG Protocol methods.
6. Flag criteria conflicts, gaps and items that require authorised standards text or programme clarification.
7. Create a document request list covering boundaries, source records, factors, calculations, uncertainty, approvals and claims.
8. Design a reporting index that links each disclosure to criterion, owner, source evidence and reviewer.
9. Peer-review the process map for missing stages and resolve one disagreement using evidence.

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Required Evidence

- GHG system process map
- Criteria crosswalk
- Document request list
- Reporting and evidence index

Acceptance Check

Every assertion is assigned to declared criteria; the process map shows all major data and approval hand-offs; the reporting index links requirements to retrievable evidence.

Activity files: activities/01-/README.md, workbook and printable PDF/checklist.*

Activity 2 — Plan a Risk-Based ISO 14064 Internal Audit

Alignment: K1 · K8 · A8 · LO1 · Duration: 45 minutes

Objective: Define audit objectives, scope, criteria, independence, risks, sampling and timetable for a two-day internal audit.

Scenario

The audit covers the 2025 organisational statement and the Lumina 400 lifecycle footprint. High-risk areas include refrigerant leaks, electricity procurement claims, imported aluminium, outsourced warehousing, supplier data from China, use-phase assumptions and a proposed '30% lower-carbon' marketing claim. The Sustainability Manager prepared several calculations and cannot audit their own work.

Materials

- Audit mandate and intended users
- Draft boundary and category totals
- Prior issues and organisational changes
- Risk and sampling workbook

Activity Questions

- What must the audit conclude?
- Which areas can materially affect the statement or user decisions?
- How will independence and competence be protected?
- Which populations and samples provide persuasive evidence?

Detailed Procedure

10. Write audit objectives that distinguish conformance, control effectiveness, data reliability and report readiness.
11. Define scope by entity, site, product, reporting period, categories, systems and locations.
12. Name criteria and record unresolved interpretation questions before fieldwork.
13. Assess inherent risk, control risk, data quality, magnitude, change and stakeholder sensitivity for each area.
14. Assign independent auditors and specialists; record conflicts and safeguards.

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15. Define populations and sample logic using high-value, unusual, high-risk and representative items.
16. Allocate time to interviews, observation, source testing, recalculation and daily team calibration.
17. Prepare interview themes and an opening-meeting agenda covering logistics, evidence access and communication.
18. Review whether planned procedures are sufficient to support each required conclusion.

Required Evidence

- Audit charter and scope
- Risk assessment
- Sampling plan
- Interview timetable and opening agenda

Acceptance Check

The plan links risks to procedures and samples, defines independence and competence, covers both statements without scope ambiguity and can support the planned conclusions.

Activity files: activities/02-/README.md, workbook and printable PDF/checklist.*

Activity 3 — Conduct Evidence Interviews and a Reporting Gap Review

Alignment: K2 · K8 · A8 · LO1 · Duration: 50 minutes

Objective: Gather, triangulate and record objective evidence, then convert confirmed gaps into clear audit findings.

Scenario

Facilities says refrigerant data comes from service reports, Finance says all invoices were supplied, Procurement used spend factors without recording versions, and Marketing already drafted a reduction claim. The report describes freight and supplier inputs as primary data although several values are estimates. One acquisition was added to 2025 totals without revisiting the base year.

Materials

- Interview role cards
- Invoices, meter extracts and service records
- Draft calculation and report extracts
- Evidence and findings workbook

Activity Questions

- Which statements require corroboration?
- What evidence is sufficient and reliable?
- Which gaps are nonconformities versus observations?
- How should sensitive emerging issues be communicated?

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Detailed Procedure

19. Prepare open, probing and confirmation questions linked to the audit criteria and risks.
20. Conduct interviews without leading the interviewee and record factual answers separately from auditor evaluation.
21. Trace selected totals from report to model, source record, unit conversion, factor and approval.
22. Corroborate interview claims through records, observation or independent data.
23. Reperform selected calculations and document any difference, cause and potential consequence.
24. Evaluate evidence against the precise criterion and determine conformity, nonconformity, OFI or insufficient evidence.
25. Write each finding as criterion, objective evidence and clear evaluation without prescribing an unsupported solution.
26. Calibrate finding wording and significance with another audit team.
27. Prepare a concise emerging-issues briefing that preserves audit independence and factual accuracy.

Required Evidence

- Interview and corroboration records
- Source-to-report audit trail
- Recalculation samples
- Finding register and issues briefing

Acceptance Check

Important interview claims are corroborated; recalculations are reproducible; findings link criterion, evidence and evaluation; fact and judgement remain clearly separated.

Activity files: activities/03-/README.md, workbook and printable PDF/checklist.*

Topic 02 — GHG Sources, Boundaries and Mitigation Activities

Alignment: K2 · K3 · K4 · K5 · A1 · A2 · A7 · LO2

Scope 1, 2 and 3 · sources, sinks and reservoirs · organisational and lifecycle boundaries · hotspots · mitigation hierarchy

Scope 1, 2 and 3 alongside ISO reporting categories

- **Scope 1:** Direct emissions from sources owned or controlled by the organisation.
- **Scope 2:** Indirect emissions from generation of purchased electricity, steam, heat or cooling.
- **Scope 3:** Other indirect upstream and downstream value-chain emissions.
- **ISO view:** Direct emissions plus five indirect-emission categories organised around imported energy, transport, products and other sources.

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Auditor application

- Classify by ownership, control and activity
- Map every source to both reporting views where needed
- Keep location- and market-based Scope 2 distinct
- Test for double counting across categories

Judgement cautions

- Outsourcing does not remove significance
- A supplier invoice is not automatically Scope 3 evidence
- Scope classification does not prove completeness
- Do not force ISO categories into GHG Protocol labels silently

Greenhouse gases, GWP and CO2e

- **Gas inventory:** Carbon dioxide, methane, nitrous oxide and covered fluorinated gases require source-specific consideration.
- **Global warming potential:** GWP converts gas mass to a common CO2-equivalent basis over a declared time horizon.
- **Source basis:** The selected IPCC assessment or programme basis must be versioned and applied consistently.
- **Audit trail:** Gas mass, factor, GWP value and CO2e result should remain separately traceable.

Auditor application

- Check all relevant gases
- Reperform factor and GWP multiplication
- Test consistency across reporting years
- Disclose changes to GWP basis

Judgement cautions

- CO2e is a convention, not measured carbon dioxide
- GWP tables change with scientific assessment
- Refrigerant blends require correct composition
- Rounding can hide material gas-level errors

Sources, sinks, reservoirs, removals and credits

- **Source:** A process or activity that releases a GHG to the atmosphere.
- **Sink:** A process that removes a GHG from the atmosphere.
- **Reservoir:** A physical unit or component that stores a GHG or carbon stock.
- **Credit:** An external unit representing a project outcome; it is not automatically an organisational removal.

Auditor application

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- Map physical processes before accounting labels
- Require monitoring for removal claims
- Evaluate reversal and permanence
- Report gross inventory and external credits separately

Judgement cautions

- Trees on site do not guarantee a quantified removal
- Avoided emissions differ from removals
- Biogenic CO2 needs declared treatment
- Purchased credits cannot erase source records

Organisational boundary decisions

- **Equity share:** Account according to the organisation's economic interest.
- **Financial control:** Include operations governed through financial and operating policies.
- **Operational control:** Include operations where the organisation can introduce and implement operating policies.
- **Evidence:** Ownership documents, leases, contracts, board rights and operating authority support the conclusion.

Auditor application

- Map every entity and operation
- Apply one consolidation approach consistently
- Review joint arrangements and leases
- Reassess after acquisitions or divestments

Judgement cautions

- Legal ownership and control can differ
- A boundary choice can change trends materially
- Contractors may remain significant indirect sources
- Do not cherry-pick low-emission operations

Reporting boundary and significant indirect emissions

- **Inventory universe:** Start with all plausible direct and indirect sources across operations and value chain.
- **Significance criteria:** Use magnitude, influence, risk, stakeholder interest, data quality and reduction opportunity.
- **Decision log:** Retain scores, evidence, inclusion decisions, exclusions, owners and review dates.
- **Change control:** Reassess significance when products, suppliers, sites, regulation or user needs change.

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Auditor application

- Challenge low-data categories
- Check procurement and logistics completeness
- Document thresholds before seeing results
- Report significant exclusions transparently

Judgement cautions

- Data difficulty is not an exclusion criterion
- Magnitude alone can miss claims risk
- Spend-based screening is not final quantification
- A significance score is judgement, not proof

Product lifecycle boundaries and assumptions

- **Goal and scope:** State intended application, audience, product configuration and decision context.
- **Functional unit:** Define the quantified function and performance basis for consistent comparison.
- **System boundary:** Specify cradle-to-gate, cradle-to-grave or other included lifecycle stages.
- **Rules:** Document cut-offs, allocation, recycled content, use profile, lifetime and end-of-life assumptions.

Auditor application

- Draw the physical process map
- Record included and excluded flows
- Test sensitivity to major assumptions
- Review comparability before benchmarking

Judgement cautions

- A product unit is not always a functional unit
- Cut-offs can hide hotspots
- Recycling can be double credited
- Use-phase assumptions often dominate results

Hotspot analysis and prioritisation

- **Magnitude:** Rank contribution by lifecycle stage, source, gas and product component.
- **Influence:** Assess control, supplier leverage, design authority and customer-use influence.
- **Confidence:** Rate data quality and test whether uncertainty could change the ranking.

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- **Decision:** Combine emissions, feasibility, cost, timing, co-impact and risk into transparent priorities.

Auditor application

- Use contribution and Pareto views
- Run sensitivity on weak high-impact inputs
- Separate screening from investment-grade analysis
- Document why lower-ranked actions proceed

Judgement cautions

- Largest is not always most controllable
- Spend hotspots can reflect price, not mass
- A precise rank can be false precision
- Do not ignore small high-risk fugitive sources

Mitigation hierarchy and audit tests

- **Avoid:** Remove unnecessary activity, demand or material before optimising it.
- **Reduce:** Improve efficiency, yield, maintenance, logistics and process control.
- **Substitute:** Change fuel, material, technology, design or service model.
- **Residual:** Treat credits or removals separately and only after credible internal and value-chain action.

Auditor application

- Quantify expected and realised abatement
- Check burden shifting across lifecycle stages
- Assign owners, milestones and evidence
- Verify claims against inventory results

Judgement cautions

- Forecast savings are not realised reductions
- Intensity can fall while absolute emissions rise
- Renewable certificates need claim controls
- Offsets are not operational reduction

Activity 4 — Define Organisational and Product Lifecycle Boundaries

Alignment: K3 · K5 · A1 · A2 · LO2 · Duration: 50 minutes

Objective: Apply consolidation, reporting-boundary and lifecycle rules to HarbourLight's organisation and Lumina 400 product.

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Scenario

HarbourLight owns the Tuas assembly plant, has a 45% stake in a Batam driver-board venture without operating authority, leases a Singapore office, uses a third-party Johor warehouse and sells panels across ASEAN. The Lumina 400 has aluminium housing, imported chips and drivers, a 50,000-hour rated life, variable regional electricity use and end-of-life recycling claims with limited evidence.

Materials

- Entity and control fact sheet
- Lumina 400 lifecycle and configuration data
- Customer-use and end-of-life assumptions
- Boundary decision workbook

Activity Questions

- Which operations are included under the selected consolidation approach?
- What function and product configuration are assessed?
- Which lifecycle stages and flows are included or excluded?
- Which assumptions could materially change the result?

Detailed Procedure

28. State the organisational-inventory purpose, product-footprint purpose, intended users and reporting period.
29. Map all entities, facilities, leases and outsourced operations with ownership and control evidence.
30. Compare equity share, financial control and operational control; select and justify one consolidation approach.
31. Apply the approach consistently and record included, excluded and unresolved organisational items.
32. Define the Lumina 400 functional unit including output, rated life and performance assumptions.
33. Map cradle-to-grave stages from raw materials through component manufacture, assembly, distribution, use and end of life.
34. Set cut-offs, allocation, recycled-content, use-profile and end-of-life rules with evidence or transparent assumptions.
35. Test boundary decisions against relevance, completeness, consistency and decision-usefulness.
36. Peer-challenge one exclusion and record the final disposition and owner.

Required Evidence

- Organisational boundary diagram
- Lifecycle process map
- Functional unit and system boundary
- Assumption and exclusion log

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Acceptance Check

Every named operation and lifecycle stage is addressed; the functional unit is comparable and measurable; exclusions and assumptions have transparent rationale, evidence and ownership.

Activity files: activities/04-/README.md, workbook and printable PDF/checklist.*

Activity 5 — Build the GHG Source, Sink and Scope Register

Alignment: K4 · K5 · A2 · A7 · LO2 · Duration: 45 minutes

Objective: Identify gases, sources, sinks and reservoirs; classify them consistently and assess significant indirect emissions.

Scenario

Potential sources include natural-gas curing ovens, emergency diesel, company vans, refrigerant leakage, purchased electricity, rooftop solar, aluminium and electronics, inbound freight, outsourced warehousing, business travel, employee commuting, product electricity use and end-of-life treatment. Facilities also claims landscaped trees and purchased credits reduce the inventory total.

Materials

- Operational and value-chain process narrative
- Energy, maintenance, procurement and logistics summaries
- Scope and ISO category prompts
- Significance and ownership workbook

Activity Questions

- Which gases and physical sources are present?
- How should each item be classified?
- Which indirect sources are significant and why?
- What evidence is needed for claimed removals or credits?

Detailed Procedure

37. Walk the organisational and product process maps and list every plausible source, sink and reservoir.
38. Identify relevant gases for combustion, refrigerants, purchased energy, materials, transport, waste and product use.
39. Classify each source into Scope 1, 2 or 3 and the applicable ISO direct or indirect-emission category.
40. Keep biogenic emissions, removals, avoided emissions and purchased credits separately identified.
41. Define significance criteria using magnitude, influence, risk, stakeholder interest, data quality and reduction opportunity.
42. Score indirect sources and record evidence and judgement supporting each score.
43. Select sources for inclusion and document transparent reasons for exclusions or provisional estimates.

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44. Assign data owner, source record, collection frequency, control and reviewer to each included source.
45. Run a completeness challenge against ledger, maintenance, procurement and waste records.

Required Evidence

- Source, sink and reservoir register
- Scope and ISO category map
- Significance assessment
- Data-owner and completeness matrix

Acceptance Check

All scenario sources and gases are addressed; Scope and ISO categories are consistent; removals and credits remain separate; significance and ownership decisions are evidence-based.

Activity files: activities/05-/README.md, workbook and printable PDF/checklist.*

Activity 6 — Diagnose Hotspots and Design a Mitigation Roadmap

Alignment: K2 · K3 · K4 · A7 · LO2 · Duration: 45 minutes

Objective: Prioritise lifecycle hotspots and propose feasible reduction approaches without burden shifting or unsupported claims.

Scenario

Screening indicates aluminium housing, product electricity use, imported electricity, air-freighted driver boards and refrigerant leakage are leading sources. Product-use estimates are uncertain, aluminium is spend-based and a redesign could reduce mass but shorten product life. Management can fund only three initiatives this year and wants a defensible customer claim.

Materials

- Screening footprint by lifecycle stage
- Data-quality ratings and uncertainty prompts
- Reduction option facts and illustrative economics
- Hotspot and roadmap workbook

Activity Questions

- Which sources remain priorities after considering influence and uncertainty?
- Could any option shift burdens or reduce product performance?
- Which actions are feasible within 24 months?
- What evidence supports future reduction claims?

Detailed Procedure

46. Rank sources by absolute emissions and lifecycle contribution.

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47. Overlay organisational influence, data confidence, trend, stakeholder importance and risk.
48. Run sensitivity checks on aluminium, product lifetime, use profile and electricity-factor assumptions.
49. Generate avoid, reduce and substitute options across design, operations, procurement, logistics, use and end of life.
50. Check every option for safety, quality, cost, feasibility, rebound, allocation and lifecycle burden shifting.
51. Estimate abatement, cost, lead time and evidence confidence without presenting forecasts as realised reductions.
52. Prioritise initiatives using transparent weighted criteria and management constraints.
53. Build 30-, 90-, 180- and 720-day milestones with owners, dependencies and decision gates.
54. Draft claim conditions linking any future statement to verified inventory evidence.

Required Evidence

- Hotspot and sensitivity dashboard
- Mitigation option register
- Prioritisation scorecard
- 24-month roadmap and claim conditions

Acceptance Check

Priorities reflect magnitude, influence and uncertainty; selected actions avoid undisclosed burden shifts; milestones, owners and evidence requirements are explicit; no forecast is presented as achieved reduction.

Activity files: activities/06-/README.md, workbook and printable PDF/checklist.*

Topic 03 — Lifecycle Assessment and Carbon Footprint Quantification

Alignment: K2 · K3 · K4 · K5 · A1 · A3 · A4 · LO3

Lifecycle phases · process mapping · lifecycle inventory · quantification models and software · inventory controls

LCA phases and product configurations

- **Goal and scope:** Define purpose, audience, functional unit, product variants and system boundary.
- **Inventory:** Compile quantified inputs and outputs for each unit process.
- **Impact assessment:** Apply declared characterisation methods such as 100-year GWP.
- **Interpretation:** Evaluate completeness, sensitivity, consistency, hotspots and limitations.

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Auditor application

- Keep configuration changes visible
- Compare products on equivalent function
- Iterate when data gaps affect the goal
- Separate measured findings from scenario results

Judgement cautions

- LCA is iterative, not a one-pass calculation
- A lighter product may perform differently
- Database completeness does not guarantee relevance
- Do not over-generalise one configuration

Process mapping for lifecycle evidence

- **Unit processes:** Break the lifecycle into decision-relevant transformation, transport, use and end-of-life stages.
- **Inputs:** Capture materials, energy, water, auxiliaries, packaging and services.
- **Outputs:** Record product, co-products, emissions, wastewater, solid waste and recyclable flows.
- **Interfaces:** Assign data owners, meters, invoices, supplier records and model links at each boundary.

Auditor application

- Walk the process with operations staff
- Reconcile mass and energy where possible
- Mark estimates and missing evidence
- Use the map to drive sampling

Judgement cautions

- A flowchart without quantities is not an inventory
- Supplier processes may be hidden
- Waste leaving site is not impact-free
- Do not mix product and organisational boundaries

Lifecycle inventory data architecture

- **Foreground data:** Company- or supplier-specific activity data for processes under direct influence.
- **Background data:** Database processes and factors for upstream or downstream activities.
- **Metadata:** Source, owner, period, geography, technology, unit, uncertainty and approval accompany each value.

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- **Balance:** Mass, energy, production and financial reconciliations reveal omissions and unit errors.

Auditor application

- Create a data dictionary
- Use stable IDs for processes and factors
- Retain raw and transformed values
- Reconcile totals to operational records

Judgement cautions

- Primary data can still be poor
- Generic data can be appropriate but must be disclosed
- Units are a frequent large-error source
- Missing metadata destroys auditability

Data hierarchy and emission-factor selection

- **Direct measurement:** Calibrated meters or continuous monitoring can support source-specific results.
- **Activity-based:** Physical activity multiplied by a relevant documented factor is common and transparent.
- **Supplier-specific:** Verified product or facility data can improve relevance when boundaries align.
- **Proxy or spend:** Screening estimates are useful when labelled and improved for material sources.

Auditor application

- Match geography, technology and vintage
- Record factor units and GWP inclusion
- Evaluate supplier boundary comparability
- Use conservative proxies where justified

Judgement cautions

- Newest is not always most representative
- A national factor may mask regional grids
- Supplier claims may net credits
- Factor precision can exceed data quality

Quantification model and calculation controls

- **Equation:** Activity data × emission factor × required conversion or GWP produces the declared result.
- **Allocation:** Shared-process burdens need a justified physical, economic or other rule.

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- **Aggregation:** Gas, source, process, stage, scope and category totals should reconcile.
- **Controls:** Locked formulas, validation rules, peer recalculation and version logs reduce silent error.

Auditor application

- Standardise units before multiplication
- Reperform high-risk rows independently
- Test sign and decimal errors
- Preserve bridges between versions

Judgement cautions

- A balanced total can contain offsetting errors
- Allocation can change conclusions
- Do not net credits into gross emissions
- Spreadsheet protection is not review

Carbon-footprint software workflow

- **Set up:** Define functional unit, product configuration, system boundary and impact method.
- **Model:** Create unit processes, enter foreground data and link controlled background datasets.
- **Analyse:** Run totals, contribution analysis, sensitivity, scenarios and data-quality review.
- **Export:** Preserve model version, datasets, settings, results and report tables for independent review.

Auditor application

- Validate one manual calculation
- Review every high-contribution model link
- Save assumptions outside the software
- Re-run after controlled corrections

Judgement cautions

- Software does not decide the boundary
- Database names can hide different technologies
- A colourful Sankey is not assurance
- Model updates can change results silently

Inventory information systems and internal controls

- **Ownership:** Every source, factor and report output has accountable preparation and review roles.

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- **Access:** Permissions separate input, method approval, review and publication responsibilities.
- **Change:** Version control records who changed what, why, when and with which approval.
- **Retention:** Source records, calculations, reviews and approvals remain retrievable for the defined period.

Auditor application

- Map data lineage end to end
- Test preventive and detective controls
- Inspect exception logs
- Sample user access and approvals

Judgement cautions

- Sustainability should not own every source
- Email approval may not control the file
- Automation can scale errors
- Backups do not prove record integrity

Base year, recalculation and verification-ready evidence

- **Base year:** Select a representative period with sufficient data and state why it is suitable.
- **Recalculation:** Define triggers for structural change, methodology change and significant error.
- **Evidence pack:** Index boundary decisions, source records, factors, calculations, controls, uncertainty and approvals.
- **Management review:** Resolve findings and accept residual limitations before the statement is issued.

Auditor application

- Publish the recalculation policy in advance
- Build an audit-ready evidence index
- Retain old and new method bridges
- Require formal inventory approval

Judgement cautions

- Ordinary performance change is not recalculation
- Acquisitions can break comparability
- A data room without an index is inefficient
- Management approval does not replace verification

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Activity 7 — Construct the Lumina 400 Lifecycle Inventory

Alignment: K3 · K4 · K5 · A1 · A3 · LO3 · Duration: 50 minutes

Objective: Build a complete, metadata-rich lifecycle inventory from the approved product boundary and configuration.

Scenario

The Lumina 400 uses an aluminium housing, glass diffuser, circuit board, LED chips, driver, wiring and packaging. Supplier data ranges from verified product footprints to generic spend estimates. Tuas assembly records electricity and scrap, regional distribution uses road and sea freight, use-phase electricity varies by market, and end-of-life recovery rates are uncertain.

Materials

- Product configuration and bill of materials
- Supplier, production, transport and use-phase records
- Background-dataset catalogue
- LCI and data-quality workbook

Activity Questions

- Which unit processes and flows are needed?
- What is foreground versus background data?
- How will metadata and data quality be captured?
- Which balances and reconciliations can reveal gaps?

Detailed Procedure

55. Confirm the functional unit, assessed configuration, lifecycle boundary and allocation rules from Activity 4.
56. Break each lifecycle stage into decision-relevant unit processes and stable process IDs.
57. List material, energy, water, transport, packaging, co-product, waste and emission flows for every process.
58. Capture quantity, unit, source, owner, period, geography, technology, collection method and approval metadata.
59. Classify each value as measured, supplier-specific, industry average, proxy, spend-based or assumption.
60. Map foreground values to controlled background factors or datasets without hiding the link.
61. Perform mass, energy, production and scrap reconciliations where data permits.
62. Rate reliability, completeness, temporal, geographical and technological representativeness.
63. Create a gap-improvement plan for high-impact weak data before final reporting.

Required Evidence

- LCI data dictionary

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- Unit-process and flow register
- Completed lifecycle inventory
- Data-quality and gap-improvement plan

Acceptance Check

Every included unit process has traceable inputs and outputs; metadata and quality ratings are complete; totals reconcile where possible; high-impact gaps have owned improvement actions.

Activity files: activities/07-/README.md, workbook and printable PDF/checklist.*

Activity 8 — Quantify and Audit the Product Carbon Footprint Model

Alignment: K2 · K4 · K5 · A3 · A4 · LO3 · Duration: 55 minutes

Objective: Use a controlled spreadsheet/software workflow to quantify the Lumina 400 footprint and test its model integrity.

Scenario

HarbourLight's current model mixes kg and tonnes, uses an undated China grid factor, links one aluminium dataset with recycled content inconsistent with procurement, and applies a 25-year product lifetime although the declared rating is 50,000 operating hours. A hidden formula omits air freight from the total, while the report rounds the final result to one decimal place.

Materials

- Activity 7 lifecycle inventory
- Illustrative factor and GWP register
- Seeded calculation model
- Software setup and audit checklist

Activity Questions

- Which model settings and datasets must be declared?
- What control failures affect the total?
- Which lifecycle stages dominate and why?
- How should corrections and residual limitations be reported?

Detailed Procedure

64. Create a controlled factor register with source, version, geography, technology, unit, GWP basis and approval.
65. Standardise activity units and document every conversion before applying factors.
66. Calculate each flow using activity data, factor and any required gas-to-CO2e conversion.
67. Aggregate results by unit process, lifecycle stage, gas and relevant scope/category.
68. Record software or workbook settings, functional unit, allocation, datasets and model version.
69. Run contribution analysis and identify high-impact model links for detailed review.

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70. Trace selected results backward to source records and forward from source to report.
71. Reperform high-risk rows, find seeded unit, factor, lifetime and formula defects, then correct them under change control.
72. Prepare an old-to-new result bridge and disclose residual estimates, uncertainty and limitations.

Required Evidence

- Controlled factor register
- Quantified product footprint
- Software/model setup record
- Audit tests and correction bridge

Acceptance Check

The model is reproducible, units and factors are controlled, seeded defects are resolved, contribution results reconcile, and the correction bridge preserves a complete audit trail.

Activity files: activities/08-/README.md, workbook and printable PDF/checklist.*

Topic 04 — Emission Calculations, Uncertainty and Performance Evaluation

Alignment: K4 · K6 · K7 · K8 · A5 · A6 · A8 · LO4

Calculations and removals · uncertainty · sensitivity · product benchmarks · trends · audit findings and follow-up

Calculating stationary, mobile, process and fugitive emissions

- **Stationary combustion:** Fuel use, calorific basis, oxidation and gas-specific factors support boiler and generator emissions.
- **Mobile combustion:** Fuel-based methods are generally stronger than distance estimates when reliable consumption data exists.
- **Process:** Material balance, chemistry or direct monitoring may quantify non-combustion releases.
- **Fugitive:** Mass-balance, service records, leak detection and equipment inventories support refrigerant and gas losses.

Auditor application

- Confirm activity units and period
- Check factor scope and oxidation assumptions
- Reconcile purchases, stocks and consumption
- Sample maintenance and loss records

Judgement cautions

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- Invoice totals may include non-reporting periods
- Distance models hide load and fuel variation
- Process factors may be technology-specific
- Top-up is not always equal to annual leakage

Purchased-energy accounting and renewable claims

- **Location-based:** Use grid-average factors reflecting physical electricity consumption in the relevant geography.
- **Market-based:** Use qualifying contractual instruments under declared quality and allocation rules.
- **Steam, heat and cooling:** Identify supplier, energy content, losses and appropriate generation factors.
- **Controls:** Check instrument ownership, geography, vintage, exclusivity, cancellation or retirement and residual mix.

Auditor application

- Report both Scope 2 views where criteria require
- Reconcile consumption to contracts
- Prevent double allocation of certificates
- Keep physical efficiency visible

Judgement cautions

- Zero-emission claims need criteria
- Certificates do not change onsite consumption
- Supplier-specific factors can embed offsets
- Renewable procurement is not the same as avoided emissions

Removals, avoided emissions and net presentation

- **Organisational removal:** A quantified atmospheric removal attributable within the declared inventory boundary.
- **Project reduction:** A change relative to a baseline under project-level criteria.
- **Avoided emissions:** A comparative consequence outside the gross inventory and dependent on a counterfactual.
- **Net claim:** A communication that must disclose gross emissions, reductions, removals, credits, boundary and period.

Auditor application

- Test ownership and boundary
- Review baseline and permanence
- Report comparative results separately
- Challenge claim wording against evidence

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Judgement cautions

- A renewable product does not create negative Scope 3 automatically
- Credits are not company removals
- Counterfactuals are uncertain
- Net numbers can conceal rising gross emissions

Uncertainty taxonomy and data-quality evaluation

- **Activity data:** Meter accuracy, estimation, completeness, temporal match and sampling affect inputs.
- **Emission factors:** Geography, technology, vintage, representativeness and statistical range affect conversion.
- **Model:** Allocation, system boundary, use profile, lifetime and end-of-life assumptions affect structure.
- **Scenario and judgement:** Future conditions, significance, exclusions and expert choices may alter conclusions.

Auditor application

- Rate reliability and representativeness
- Separate known error from uncertainty
- Prioritise improvement by decision impact
- Disclose residual uncertainty

Judgement cautions

- A single percentage can hide different uncertainty types
- Low uncertainty does not mean no bias
- Qualitative assessment can still be rigorous
- Do not claim confidence beyond evidence

Sensitivity, scenarios and quantitative uncertainty

- **Sensitivity:** Change one influential input or assumption to test its effect on the result.
- **Scenario:** Compare coherent alternative futures, technologies, boundaries or use profiles.
- **Propagation:** Combine input distributions analytically or through Monte Carlo simulation where justified.
- **Decision threshold:** Ask whether uncertainty could change ranking, conformance, target status or investment choice.

Auditor application

- Start with high-impact weak inputs
- Define plausible ranges and sources
- Report distributions and assumptions

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- Tie analysis to a decision

Judgement cautions

- Wide ranges are not a substitute for better data
- Inputs may be correlated
- Monte Carlo precision can be misleading
- Scenario outcomes are not forecasts

Product carbon-footprint benchmarking

- **Comparable unit:** Align functional unit, performance, service life and product configuration.
- **Comparable boundary:** Match lifecycle stages, cut-offs, allocation, recycling and use assumptions.
- **Comparable method:** Align GWP basis, database, geography, technology and assessment vintage.
- **Evidence source:** Use verified EPDs, product category rules, industry datasets and peer-reviewed studies with stated limitations.

Auditor application

- Build a comparability screen before ranking
- Normalise only when defensible
- Show range and quartile, not one magic number
- Verify claims before marketing use

Judgement cautions

- Different PCRs may prevent comparison
- Regional grid differences can dominate
- Best-in-class data may be selective
- A lower number does not prove overall environmental superiority

Performance over time and causal interpretation

- **Absolute:** Total emissions show atmospheric contribution and scale effects.
- **Intensity:** Emissions per unit of output or function show efficiency but depend on denominator quality.
- **Bridge:** Separate activity, structure, factor, method and genuine reduction effects between periods.
- **Control chart:** Monitor trends, thresholds and anomalies without confusing normal variation with improvement.

Auditor application

- Recalculate base years when policy triggers
- Reconcile initiatives to inventory sources

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- Track data-quality improvement
- Investigate unexpected favourable variance

Judgement cautions

- Intensity can improve while absolute emissions rise
- Grid decarbonisation is not an internal project
- Method changes can create artificial trends
- Targets do not prove achievement

Audit findings, closing meeting and follow-up

- **Conformity:** Evidence demonstrates that the stated criterion is met.
- **Nonconformity:** Objective evidence shows a requirement is not fulfilled.
- **Observation or OFI:** A risk or improvement insight that is not mislabelled as a requirement breach.
- **Closure:** Correction addresses the immediate issue; corrective action addresses cause and effectiveness.

Auditor application

- Write criterion-evidence-evaluation findings
- Calibrate severity before closing
- Invite factual clarification without negotiating facts
- Verify implementation and effectiveness separately

Judgement cautions

- Do not hide requirements inside recommendations
- A missing record is not automatically a wrong result
- Root cause is not repeat the finding
- Closing evidence can be superficial

Activity 9 — Evaluate Uncertainty and Product Benchmark Performance

Alignment: K6 · K7 · A5 · A6 · LO4 · Duration: 45 minutes

Objective: Determine material uncertainty factors and compare the Lumina 400 with credible, method-aligned benchmarks.

Scenario

The corrected Lumina 400 result is 186 kg CO2e per functional unit. Supplier aluminium data varies by ±25%, China grid factors differ by region, customer operating hours range widely and end-of-life recycling evidence is weak. Marketing compares the result with three competitor declarations: 175, 210 and 245 kg CO2e, but their functional units, lifetimes, boundaries and PCR editions differ.

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Materials

- Activity 8 model and contribution results
- Input ranges and data-quality ratings
- Three fictional competitor EPD extracts
- Uncertainty and benchmark workbook

Activity Questions

- Which uncertainty factors could change the decision?
- What sensitivity or scenario tests are proportionate?
- Are the competitor results comparable?
- What claim, if any, can HarbourLight support?

Detailed Procedure

73. Identify activity-data, factor, model, sampling, scenario and boundary-judgement uncertainty.
74. Rate likelihood, range, direction, evidence quality and decision impact for each factor.
75. Select high-impact weak inputs for sensitivity testing and document plausible ranges and sources.
76. Run one-at-a-time sensitivities and coherent scenarios for aluminium, grid, lifetime and recycling.
77. Determine whether ranges change hotspot ranking, total result, target status or proposed claim.
78. Screen each benchmark for functional unit, product performance, system boundary, allocation, geography, technology, GWP basis and vintage.
79. Exclude non-comparable records or normalise only where a defensible method and evidence exist.
80. Present ranges, comparability limitations and ranking without false precision.
81. Recommend approve, qualify, defer or reject the proposed product claim and state evidence needed for reconsideration.

Required Evidence

- Uncertainty register
- Sensitivity and scenario results
- Benchmark comparability screen
- Performance and claim recommendation

Acceptance Check

All material uncertainty classes are addressed; sensitivity links to decisions; only genuinely comparable benchmarks influence the conclusion; claim wording matches evidence and limitations.

Activity files: activities/09-/README.md, workbook and printable PDF/checklist.*

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Activity 10 — Close the Audit and Issue a Verification-Ready Report

Alignment: K1 · K8 · A1 · A5 · A6 · A8 · LO1 · LO4 · Duration: 40 minutes

Objective: Consolidate evidence, communicate balanced conclusions and verify corrective action for the ISO 14064 internal audit.

Scenario

Fieldwork confirms strong electricity reconciliation and management commitment, but identifies an unapproved factor change, omission of refrigerant leakage, weak supplier-data metadata, an inconsistent product lifetime, incomplete uncertainty disclosure and an unsupported 30% lower-carbon claim. Management disputes one finding and asks the audit team to prescribe exact corrective formulas before the closing meeting.

Materials

- Outputs from Activities 1-9
- Consolidated evidence and finding candidates
- Closing-meeting role cards
- Report and corrective-action workbook

Activity Questions

- Which findings are supported and how significant are they?
- How should disagreement be handled without weakening evidence?
- What belongs in the audit conclusion and report?
- How will correction, cause and effectiveness be verified?

Detailed Procedure

82. Consolidate audit trails, recalculations, boundary decisions, uncertainty and benchmark evidence from Activities 1-9.
83. Confirm each finding's criterion, objective evidence, evaluation and consequence; remove duplicates.
84. Calibrate significance using magnitude, pervasiveness, control failure, user impact and claims risk.
85. Prepare a balanced closing presentation covering objectives, scope, positive practices, limitations, findings and next steps.
86. Conduct a simulated closing meeting, invite factual clarification and document unresolved disagreement.
87. Draft the internal-audit report with executive summary, criteria, methods, samples, findings, conclusion and distribution control.
88. Require management to define correction, cause analysis, corrective action, owner, due date and completion evidence.
89. Plan follow-up tests that distinguish implementation from sustained effectiveness.
90. Issue a verification-readiness brief stating resolved items, residual limitations and conditions before external use.

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Required Evidence

- Final finding register
- Closing presentation and meeting record
- ISO 14064 internal-audit report
- Corrective-action and follow-up plan

Acceptance Check

Every conclusion is evidence-based; the report is balanced and scope-accurate; disagreement is transparent; corrective action addresses cause; follow-up tests effectiveness rather than document completion alone.

Activity files: activities/10-/README.md, workbook and printable PDF/checklist.*

Assessment Preparation

Written Assessment (WA-SAQ): 8 open-ended questions covering K1-K8, 1 hour, open book.

Case Study (CS): one coherent scenario with 4 open-ended questions covering A1-A8, 1 hour, open book.

Competency requires a Competent result for every assessed K and A criterion and at least 75% attendance.

Sources and Further Reading

- Tertiary course page: https://www.tertiarycourses.com.sg/wsq-iso-14064-internal-auditor-training.html
- ISO 14064 overview: https://en.wikipedia.org/wiki/ISO_14064
- Seedling ISO 14064 verification guide: https://www.seedling.earth/post/iso-14064-verification-the-ultimate-guide
- Greenly ISO 14064 objectives and requirements: https://greenly.earth/en-us/blog/company-guide/iso-14064-objectives-and-requirements
- LRQA ISO 14064 overview: https://www.lrqa.com/en-gb/iso-14064/
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- DNV guide to becoming an ISO internal auditor: https://www.dnv.com/assurance/articles/how-to-become-an-iso-internal-auditor/
- DNV guide to ISO audits: https://www.dnv.com/assurance/articles/what-is-an-iso-audit/
- ISO 14064-1:2018 official catalogue: https://www.iso.org/standard/66453.html
- ISO 14064-3:2019 official catalogue: https://www.iso.org/standard/66455.html
- ISO 19011:2018 official catalogue: https://www.iso.org/standard/70017.html
- GHG Protocol Corporate Standard: https://ghgprotocol.org/corporate-standard

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- GHG Protocol Scope 3 Standard: https://ghgprotocol.org/corporate-value-chain-scope-3-standard
- IPCC Sixth Assessment Report: https://www.ipcc.ch/assessment-report/ar6/

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